SYSTEM FOR AN OPTICALLY-NATIVE CENTRAL PROCESSING UNIT

Reinaldo Javier Aguiar Marcano

March 11, 2026

Title: SYSTEM FOR AN OPTICALLY-NATIVE CENTRAL PROCESSING UNIT

Inventor: Reinaldo Javier Aguiar Marcano, a citizen of the United States, residing at 2302 Britton Ridge Drive, Katy, TX 77494.

Cross-Reference to Related Applications

This application claims the benefit of and priority to U.S. Patent Application No. 19/559,705, the contents of which are hereby incorporated by reference in their entirety.

Abstract

A system for a Central Processing Unit (CPU) that is natively designed to process ultra-wide data words received directly from an integrated, massively parallel optical interface. The system comprises an execution core with arithmetic logic units (ALUs), registers, and instruction decoders specifically architected to operate on data words of 256 bits or greater. This execution core is directly coupled to an optical I/O unit on the same die, eliminating the need for traditional serial-to-parallel conversion logic and other bottlenecks associated with electrical I/O. This optically-native design allows the CPU to ingest and process data at the full line rate of an ultra-high-bandwidth optical bus, enabling unprecedented computational throughput.

Background of the Invention

Traditional CPU cores are designed around the limitations of their electrical I/O pins. They ingest data in relatively narrow chunks (e.g., 64 bits at a time) and often require complex serial-to-parallel conversion logic (SERDES) to handle high-speed electrical communication. The internal architecture of the CPU, including the width of its registers and ALUs, is thus a reflection of these I/O constraints. Even if a faster external bus were available, the CPU core itself would not be able to take advantage of it, as it is not designed to handle a massively wide data word in a single clock cycle.

Summary of the Invention

The present invention is a system for an optically-native CPU, which is a processor core architected from the ground up to leverage the capabilities of a massively parallel optical interface. The system integrates the processing logic and the optical I/O into a single, cohesive unit, removing the bottlenecks that separate these functions in traditional designs.

The system comprises two main components on a single semiconductor die: an optical I/O unit, consisting of an array of photon sensors and emitters, and an execution core. The execution core is designed with an ultra-wide internal architecture. Its internal data paths, registers, and the operands for its arithmetic logic units (ALUs) are designed to match the width of the optical bus, for example, 256 bits, 1024 bits, or wider. When an optical data frame arrives, the ultra-wide data word is passed directly to the execution core's registers without serial-to-parallel conversion. This allows the core to perform a computational operation on the entire 1024-bit data word in a single instruction, achieving a level of computational density and performance impossible with conventional CPU designs.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government and its military and defense agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental, defense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.

Brief Description of the Drawings

FIG. 1 is a block diagram of the optically-native CPU system, illustrating the direct coupling of the optical I/O unit and the wide-architecture execution core.

Detailed Description of the Invention

Referring to **FIG. 1**, the system for an optically-native CPU (100) is shown. The system is integrated on a single die. It includes an optical I/O unit (110), comprising an array of photon sensors and emitters that form the physical interface to an optical bus. The optical I/O unit is directly coupled to an ultra-wide execution core (120). Within the execution core, the key components are shown with wide data paths, including a set of ultra-wide registers (121) and an Arithmetic Logic Unit (ALU) (122) configured to operate on operands with a bit-width matching the optical bus (e.g., ¿= 256 bits). An incoming ultra-wide data word (130) from the optical bus is placed directly into a register and processed by the ALU in a single operation. This design eliminates the traditional I/O bottleneck.

Claims

What is claimed is:

1. A central processing unit system, comprising:
    - a massively parallel optical input/output (I/O) unit integrated on a semiconductor die; and
    - an execution core integrated on the same semiconductor die and directly coupled to the optical I/O unit;
    - wherein the execution core comprises at least one register and at least one arithmetic logic unit (ALU) configured to operate on data words with a bit-width greater than 128 bits.
2. The system of claim 1, wherein the bit-width is 256 bits or greater.
3. The system of claim 1, wherein the bit-width is 1024 bits or greater.
4. The system of claim 1, wherein the system is configured to receive an data word from the optical I/O unit and load it into the register in a single clock cycle, without serial-to-parallel conversion.

5. The system of claim 1, wherein the optical I/O unit is an implementation of the interface described in U.S. Patent Application No. 19/559,705.

Drawings

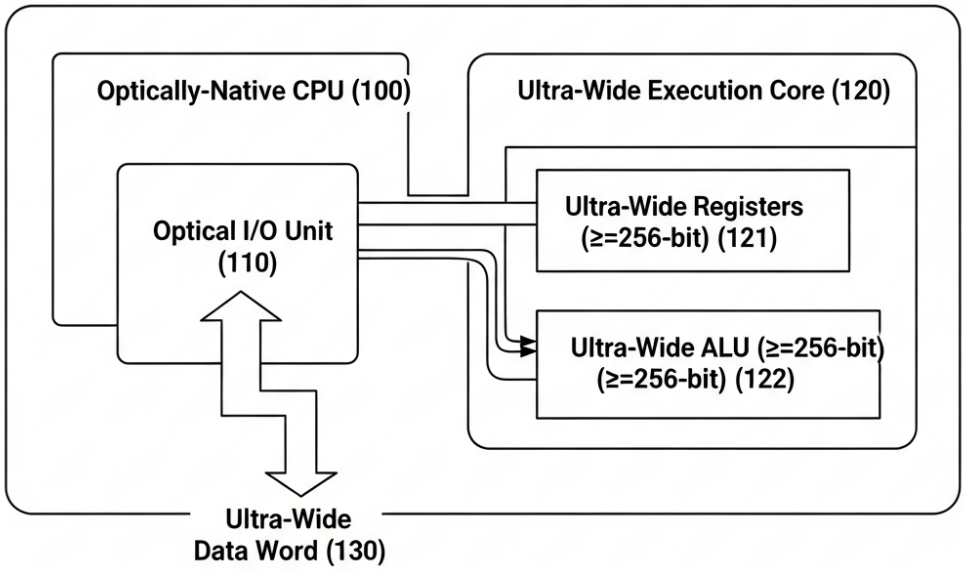


Figure 1: Block diagram of the optically-native CPU system, illustrating the direct coupling of the optical I/O unit and the wide-architecture execution core.